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DATABASE MANAGEMENT SYSTEMS
PROJECT REPORT

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HR Data Dashboard Automation

Automated Employee Sentiment Analysis Using Google Gemini and n8n: An AI-Driven Human Resources Feedback Processing System

Abstract—This paper presents an automated employee sentiment analysis system developed using the n8n workflow automation platform, Google Sheets, and the Google Gemini large language model (LLM). The system continuously monitors an HR feedback spreadsheet named IK_Analiz_Sistemi and triggers an AI-driven sentiment analysis pipeline whenever a new employee feedback entry is recorded. The Gemini model processes each employee's free-text feedback as an HR specialist, generating a one-sentence summary of motivation level and recommended actions. The resulting sentiment output is automatically written back to the spreadsheet under the AI_DUYGU_ANALIZI column. The proposed system demonstrates a practical and scalable approach to augmenting human resources decision-making with generative AI, requiring no programming expertise from HR practitioners.

Keywords—sentiment analysis, human resources, workflow automation, large language model, Google Gemini, employee feedback, n8n, organizational behavior

I. INTRODUCTION

Employee feedback is a vital source of organizational intelligence. When processed effectively, it reveals patterns in workforce motivation, identifies early signs of burnout, and informs managerial interventions before minor dissatisfaction escalates into turnover [1]. However, the volume of feedback in medium and large organizations frequently exceeds the capacity of HR teams to analyze it manually in a timely manner.

Recent advances in large language models (LLMs) have demonstrated strong capabilities in affective text analysis, including the detection of sentiment polarity, emotion intensity, and actionable signals in natural language [2]. Simultaneously, low-code workflow automation platforms such as n8n have matured to the point where non-programmers can compose complex data pipelines by connecting pre-built nodes.

This paper describes the design and implementation of an end-to-end HR feedback analysis system that combines these two technologies. When an employee submits a feedback entry in a shared Google Sheets workbook, the system automatically extracts the text, forwards it to Google Gemini with a structured HR-specialist prompt, and writes the resulting sentiment summary back to the sheet. The process is fully automated and completes within seconds of data entry.

The paper is structured as follows. Section II reviews relevant literature on automated sentiment analysis and HR analytics. Section III describes the overall system architecture. Section IV details the three-node n8n workflow implementation. Section V discusses results and limitations. Section VI concludes the paper.

II. RELATED WORK

A. Sentiment Analysis in Organizational Settings

Sentiment analysis, also termed opinion mining, has been applied to employee-generated text across several domains including exit interviews, engagement surveys, and internal communication channels [3]. Early approaches relied on lexicon-based methods that matched words against polarity dictionaries. Machine-learning classifiers trained on labeled review corpora subsequently improved accuracy, particularly for domain-specific language [4].

The introduction of transformer-based pre-trained models such as BERT and its successors marked a step change in sentiment analysis performance [5]. These models capture contextual nuance that keyword-based methods miss, enabling finer-grained classification of emotion categories such as frustration, enthusiasm, and disengagement.

B. LLMs for Human Resources Applications

Instruction-following LLMs have recently been applied to HR tasks including job description generation, resume screening, and employee communication drafting [6]. Their ability to follow complex natural-language instructions makes them particularly well-suited to open-ended analytical tasks where the desired output format can be specified in plain text rather than programmed explicitly.

Several studies have examined the reliability of LLM-generated sentiment labels relative to human annotators. Findings suggest high agreement rates on clearly valenced text, with divergence primarily occurring in ambiguous or culturally specific language [7]. This limitation is especially relevant for multilingual deployments, as discussed in Section V.

C. Workflow Automation in HR

HR automation research has historically focused on structured data processing such as payroll computation and leave management [8]. The emergence of low-code integration platforms has expanded automation reach to semi-structured and unstructured data workflows. Recent industrial case studies report substantial reductions in manual processing time when routine HR tasks are delegated to automated pipelines [9].

III. SYSTEM ARCHITECTURE

The proposed system follows a three-tier event-driven architecture. The data tier is a Google Sheets workbook serving as both the input interface for employee feedback and the output store for AI-generated sentiment summaries. The processing tier is the n8n workflow engine, which orchestrates data flow and manages API interactions. The inference tier is the Google Gemini LLM, which performs the sentiment analysis task.

TABLE I. System Component Overview

Tier	Technology	Function
Data	Google Sheets (IK_Analiz_Sistemi)	Employee feedback input and AI output storage
Orchestration	n8n Workflow Engine	Event detection, node execution, API management
Inference	Google Gemini (gemini-3-flash-preview)	LLM-based sentiment and motivation analysis
Auth — Input	Google Sheets Trigger OAuth 2.0	Secure polling access to the spreadsheet
Auth — Output	Google Service Account	Write-back access for AI analysis results

The spreadsheet schema contains four columns: ISIM (employee name, used as the matching key), DEPARTMAN (department), GERI_BILDIRIM (free-text feedback), and AI_DUYGU_ANALIZI (AI-generated sentiment summary). The workflow is configured as active and executes in n8n's v1 execution order with binary data in separate mode.

IV. WORKFLOW IMPLEMENTATION

A. Trigger Node: Google Sheets Trigger

The first node is an n8n Google Sheets Trigger (typeVersion 1) configured to poll the IK_Analiz_Sistemi workbook (document ID: 12lWhcGjy5UB6vgqCbHWKluTmkdPpOWgvvl-JQiWcd-4) every minute. It monitors Sheet1 (gid=0) for the rowAdded event. Authentication uses OAuth 2.0 via the Google Sheets Trigger account credential. Each detected new row immediately initiates a downstream execution chain, passing the full row JSON to the next node.

B. Inference Node: Google Gemini

The second node uses the n8n LangChain Google Gemini integration (typeVersion 1.2) with the gemini-3-flash-preview model. The prompt is dynamically constructed by inserting the GERI_BILDIRIM field value from the triggering row. The instruction, translated from Turkish, reads: "Read the following

employee feedback and perform sentiment analysis as an HR expert. Summarize the motivation status and recommended action in a maximum of one sentence: [feedback text]."

This prompt design reflects established best practices for LLM-based sentiment analysis: it specifies the evaluator persona (HR expert), defines the analytical task (sentiment analysis), constrains the output length (one sentence), and requests both an assessment dimension (motivation status) and an action dimension (recommended action) [10]. No additional tools or function-calling configurations are enabled for this node.

C. Write-Back Node: Google Sheets Append/Update

The third node uses the n8n Google Sheets node (typeVersion 4.7) in appendOrUpdate mode, authenticated via a Google Service Account rather than OAuth 2.0. This choice of credential type is significant: service accounts are preferable for server-side write operations as they do not expire with user session tokens. The node matches on the ISIM column to locate the correct row and writes the Gemini response, extracted from the JSON path `content.parts[0].text`, to the AI_DUYGU_ANALIZI column. Only these two fields are mapped, leaving all other columns untouched.

V. DISCUSSION

A. Practical Advantages

The proposed system offers several concrete benefits for HR operations. First, it eliminates the latency between feedback submission and sentiment assessment: results appear in the spreadsheet within seconds of data entry, enabling HR managers to act on emerging issues without waiting for periodic batch reports. Second, because Google Sheets serves as the interface layer, neither employees submitting feedback nor managers reading analyses require any new software or training. Third, the use of a Service Account for write operations decouples the automation from any individual user's Google session, making the pipeline robust to credential rotation and organizational restructuring.

B. Comparison with Previous Workflow

This system shares architectural patterns with a previously implemented patient intake automation workflow that also used Google Sheets, n8n, and Google Gemini [cf. prior work]. Key differences include the domain (HR versus clinical), the prompt strategy (single-sentence summary versus risk evaluation), and the authentication mechanism for write-back operations (Service Account versus OAuth 2.0). The HR system's active deployment state contrasts with the patient intake workflow's inactive state, suggesting it is in live production use.

C. Limitations

Several limitations merit consideration. The one-minute polling interval introduces a maximum latency of sixty seconds between feedback submission and AI processing; a webhook-triggered architecture would eliminate this delay. The prompt is written in Turkish, limiting the system's applicability to Turkish-language feedback without modification. The one-sentence output constraint, while promoting conciseness, may lose nuance for complex or multi-dimensional feedback entries.

From a data governance perspective, employee feedback is sensitive personal data. Transmitting it to a third-party LLM API raises privacy considerations under regulations such as GDPR. Organizations deploying this system should conduct a data protection impact assessment and consider on-premises LLM alternatives for high-sensitivity deployments.

D. Future Directions

Planned extensions include: (1) a multi-label sentiment scheme that categorizes feedback into dimensions such as workload, management, compensation, and career growth; (2) a dashboard layer that aggregates AI_DUYGU_ANALIZI entries by department and trend over time; (3) an escalation sub-workflow that automatically notifies a designated HR officer when highly negative sentiment is detected; and (4) prompt localization to support multilingual feedback in global organizations.

VI. CONCLUSION

This paper presented an automated employee sentiment analysis system that integrates the n8n workflow engine, Google Sheets, and the Google Gemini LLM into a cohesive, low-code pipeline. The system continuously monitors an HR feedback workbook, applies AI-driven sentiment analysis to each new entry, and persists the result without manual intervention. The architecture is lightweight, extensible, and deployable by HR practitioners without programming expertise. By making sentiment intelligence available in near real time, the system enables more responsive and data-informed human resources management.

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